Introduction to Java

* Why was Java Created?
→ before java, C & C++ dominated, but it has Major problems
: 1] Portability issue

Code → Compile → Platform 1 ($O.S_1$ + $Processor_1$) → Machine code are different after compilation for diffrn platforms.

Platform 2 ($O.S_2$ + $Processor_2$) → Machine code are different after compilation for diffrn platforms.

2] Complexity
→ features like pointers, manual memory allocation & multiple inheritance made C++ complex.

3] Security
→ C++ programs allowed unsafe operations, making systems vulnerable

* Platform
→ Platform is a combination of : • Processor
• Operating system

- Each platform has a unique Instruction Set Architecture (ISA) which defines how software commands are executed at the processor level. This difference in ISA causes compiled binaries to be platform specific

* Java's Solution to these problems

- Java was designed to address the three main issues

1] Portability: Bytecode + JVM was solution

→ Java source code is compiled into platform-independent bytecode (.class files). JVMs on each platform interpret this bytecode into platform-specific machine code at runtime

2] Simplicity: Removal of complex C++ features

→ Java eliminated pointers, multiple inheritance (via classes) and manual memory management to simplify programming.

3] Security: JVM's sandbox environment

→ JVM runs bytecode in restricted environment preventing unsafe operations, protecting the system from malicious code

* Points to be Noted

- While bytecode is independent of platform, JVM is platform dependent meaning a JVM must be implemented for each platform.

→ Java was used extensively both on the backend (via servlets) and the frontend (via applets, though applets are now deprecated).

- Java follows WORA (Write once, Run Anywhere)